THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Parallel and Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If co-interior angles add to 180°, the lines are: 1

(a) intersecting **(b)** perpendicular **(c)** parallel **(d)** unequal

2. Angle $A = 64^\circ$ and angle B are vertically opposite. Angle B is: 1

(a) 74° **(b)** 54° **(c)** 84° **(d)** 64°

3. An angle of 90° is: 1

(a) right **(b)** acute **(c)** obtuse **(d)** reflex

4. The sum of angles around a point is 360°. If three angles are 120°, 70°, 60° the fourth is: 1

(a) 120° **(b)** 130° **(c)** 100° **(d)** 110°

5. If x and 51° are complementary, $x =$ 1

(a) 29° **(b)** 49° **(c)** 39° **(d)** 59°

6. In a linear pair, one angle is 5 times the other. The smaller angle is: 1

(a) 20° **(b)** 40° **(c)** 30° **(d)** 50°

7. Two parallel lines are cut by a transversal. If an interior angle is 112°, the co-interior angle on the same side is: 1

(a) 88° **(b)** 68° **(c)** 78° **(d)** 58°

8. If alternate interior angles are not equal, the lines are: 1

(a) not parallel **(b)** parallel **(c)** perpendicular **(d)** coincident

9. Angle $A = 97^\circ$ and angle B are vertically opposite. Angle B is: 1

(a) 117° **(b)** 87° **(c)** 107° **(d)** 97°

10. An angle of 125° is: 1

(a) reflex **(b)** right **(c)** obtuse **(d)** acute

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Lines $l \parallel m$ and a transversal. A pair of alternate interior angles are $(x + 25)^\circ$ and 80°. Find x. 2

12. Two angles are supplementary and one is 3 times the other. Find them. 2

13. Find x if $(2x + 30)^\circ$ and (80°) form a linear pair. 2

14. Lines $l \parallel m$ and a transversal. A pair of alternate interior angles are $(x + 15)^\circ$ and 110°. Find x. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 15)^\circ$ and its corresponding angle $= (2x + 60)^\circ$. Find x and the angle. 3

16. Co-interior angles on parallel lines are $(4x + 12)^\circ$ and $(2x + 18)^\circ$. Find x and both angles. 3

17. Two lines intersect forming angles $(5x - 8)^\circ$ and $(3x + 18)^\circ$ as vertically opposite angles. Find x and all four angles. 3

OR

Lines $l \parallel m$, transversal t. Angle $\angle 1 = (3x + 8)^\circ$ and its corresponding angle $= (2x + 45)^\circ$. Find x and the angle. 3

18. Co-interior angles on parallel lines are $(4x + 4)^\circ$ and $(2x + 14)^\circ$. Find x and both angles. 3

OR

Two lines intersect forming angles $(5x - 12)^\circ$ and $(3x + 30)^\circ$ as vertically opposite angles. Find x and all four angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 75^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 75°. How many of the eight angles measure 75° and how many measure 105°? Give reasons. 5

20. Lines $l \parallel m$ and transversal t. Angles at l: $\angle 1 = 110^\circ$. Write the values of the corresponding, alternate interior, vertically opposite and co-interior angles of $\angle 1$ at m, with reasons. 5

OR

Two parallel lines are cut by a transversal and one of the eight angles is 110°. How many of the eight angles measure 110° and how many measure 70°? Give reasons. 5

END OF QUESTION PAPER — Total: 40 Marks

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